



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	ELECTRODYNAMICS III				
Type of Course	Core in Major				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Electrodynamics I and II				
Course Summary	Students explore the intricate theoretical foundations and advanced applications of electromagnetism, delving into topics such as Maxwell's equations, electromagnetic waves, electromagnetic field theory, relativistic electrodynamics, and their applications in modern physics and engineering.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate mastery of Maxwell's equations and their applications in various contexts	Applying	Procedural Knowledge	Written exams, problem sets

CO2	Analyze electromagnetic wave propagation and interaction with matter using advanced mathematical techniques	Analyzing	Procedural Knowledge	Homework assignments, exams
CO3	Explain the physical significance of electromagnetic potentials and gauge transformations	Understanding	Conceptual Knowledge	Class discussions, presentations
CO4	Predict and interpret the behavior of electromagnetic fields in complex geometries and boundary conditions	Analyzing	Procedural Knowledge	Laboratory experiments, simulations
CO5	Apply relativistic electrodynamics principles to describe electromagnetic phenomena in the context of special relativity	Applying	Procedural Knowledge	Projects, research papers
CO6	Design and analyze advanced electromagnetic systems and devices, demonstrating creative problem-solving skills	Creating	Procedural Knowledge	Design projects, presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	CONSERVATION LAWS AND ELECTROMAGNETIC WAVES		12	18
	1	The Continuity Equation	1	

	2	Poynting's Theorem	1	
	3	Newton's Third Law in Electrodynamics	1	
	4	Conservation of Momentum, Angular Momentum	2	
	5	Magnetic Forces Do No Work	1	
	6	Electromagnetic Waves in Matter – Reflection and Transmission (Normal and Oblique Incidence)	3	
	7	Electromagnetic Waves in Conductors; Reflection at a Conducting Surface	1	
	8	The Frequency Dependence of Permittivity	2	
	Sections 8.1.1, 8.1.2, 8.2.1, 8.2.3, 8.2.4, 8.3, 9.3.2, 9.3.3, 9.4.1 – 9.4.3 of Book 1			
II	POTENTIALS AND FIELDS		12	18
	9	Scalar and Vector Potentials; Gauge Transformations	2	
	10	Coulomb and Lorenz Gauge; Lorentz Force Law in Potential Form	2	
	11	Retarded Potentials, Jefimenko's Equations	2	
	12	The Lienard – Wiechert Potentials; Fields of a Moving Point Charge	3	
	13	Multipole Expansion of the Scalar and Vector Potentials	3	
	Sections 10.1 – 10.3, 3.4.1 – 3.4.4, 5.4.3 of Book 1			
III	RADIATION		8	12
	14	What is Radiation; Electric Dipole Radiation	3	
	15	Magnetic Dipole Radiation; Radiation from an Arbitrary Source	3	
	16	Power Radiated by a Point Charge – Larmor and Lienard Formulae	2	

	Sections 11.1.1 – 11.2.2 of chapter 11 of Book 1			
IV	ELECTRODYNAMICS AND RELATIVITY		16	22
	17	Review of Special Theory of Relativity	2	
	18	The Structure of Space-Time; Relativistic Mechanics – Proper Time and Proper Velocity, Relativistic Energy and Momentum, Relativistic Dynamics	4	
	19	Magnetism as a Relativistic Phenomenon	2	
	20	How the Fields Transform	2	
	21	The Field Tensor	2	
	22	Electrodynamics in Tensor Notation	2	
	23	Relativistic Potentials	2	
	Review of sections 12.1.1 – 12.1.3; sections 12.1.4, 12.2.1, 12.2.2, 12.2.4 , 12.3.1 – 12.3.5 of chapter 12 of Book 1			
V	Open-Ended Module: WAVEGUIDES AND TRANSMISSION LINES		12	

Books and References:

1. Introduction to Electrodynamics, 5th Edn. by David J Griffiths; Prentice Hall India Learning Pvt. Ltd (Book 1)
2. Classical Electrodynamics by John David Jackson
3. Electrodynamics by Georgi V. Shilov
4. Principles of Electrodynamics by Melvin Schwartz
5. Electromagnetic Fields and Waves by Vladimir Rojansky
6. Electromagnetic Waves by David H. Staelin, Ann W. Morgenthaler, and Jin Au Kong

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	3	1	2	1	3	1	1	1	3	2	1

CO 2	3	3	3	1	2	1	3	1	1	1	3	2	1
CO 3	3	1	2	0	3	1	3	1	1	0	3	2	1
CO 4	3	3	3	2	2	2	3	1	1	1	3	2	1
CO 5	3	2	2	1	3	1	3	1	1	1	3	2	2
CO 6	2	2	3	3	1	3	3	1	2	1	3	2	2

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	